

Task 60: Course Schedule

Problem Description

There are a total of `numCourses` courses you have to take, labeled from 0 to `numCourses - 1`. You are given an array `prerequisites` where `prerequisites[i] = [ai, bi]` indicates that you must take course `bi` first if you want to take course `ai`. Return `true` if you can finish all courses. Otherwise, return `false`.

Java Solution

```
class Solution {
    public boolean canFinish(int numCourses, int[][] prerequisites) {
        int[] indegree = new int[numCourses];
        List<List<Integer>> adj = new ArrayList<>();
        for (int i = 0; i < numCourses; i++) {
            adj.add(new ArrayList<>());
        }

        for (int[] pre : prerequisites) {
            adj.get(pre[1]).add(pre[0]);
            indegree[pre[0]]++;
        }

        Queue<Integer> q = new LinkedList<>();
        for (int i = 0; i < numCourses; i++) {
            if (indegree[i] == 0) q.offer(i);
        }

        int count = 0;
        while (!q.isEmpty()) {
            int curr = q.poll();
            count++;
            for (int next : adj.get(curr)) {
                indegree[next]--;
                if (indegree[next] == 0) {
                    q.offer(next);
                }
            }
        }

        return count == numCourses;
    }
}
```